

063-0605-22L

Computational Structural Design I

Computational Graphic Statics

Dr. Lluís Enríque

Block Research Group (Prof. Dr. Philippe Block)

HS2022



BRG

<http://block.arch.ethz.ch>



@blockresearchgroup



BRG

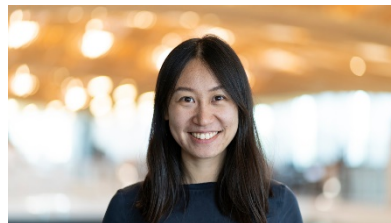


Lecturer

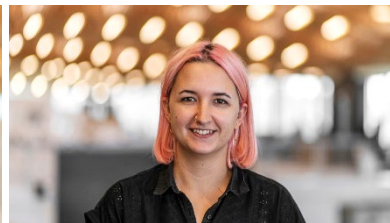


Dr. Lluís Enrique

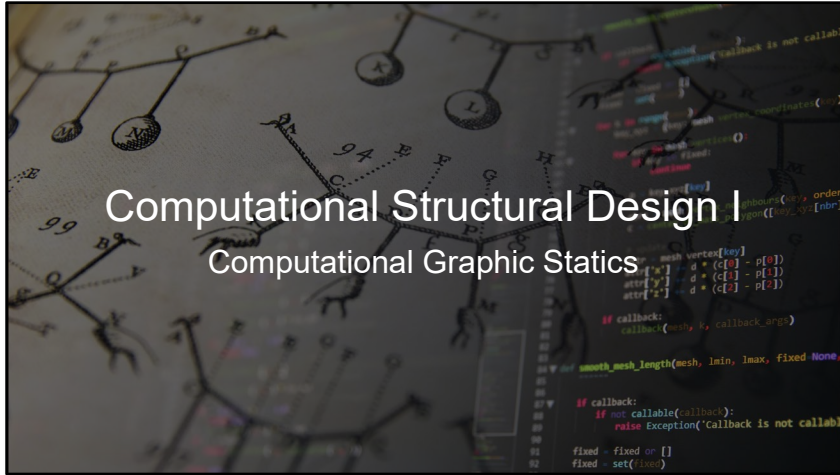
PhD student instructors



Chaoyu Du



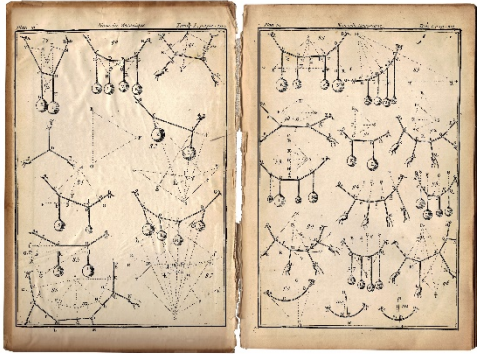
Selina Bitting



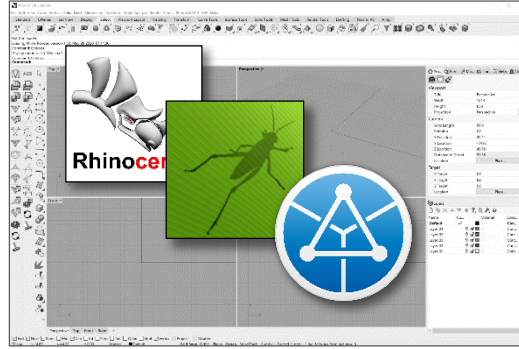
HS2022



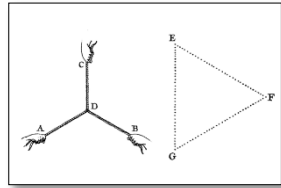
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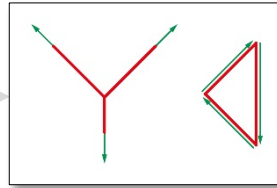
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Explore new **structural design opportunities**
by combining **graphic statics** with **computational tools**

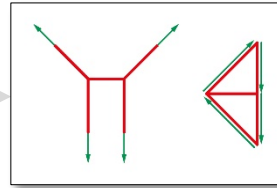


Introduction



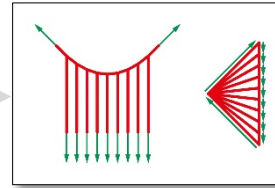
Module I

Procedural Graphic Statics I (single node)



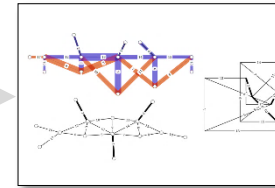
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Procedural Graphic Statics II (multi node)



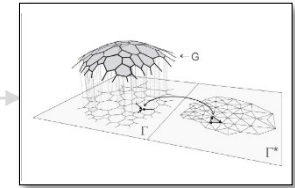
Module III

Programming Graphic Statics



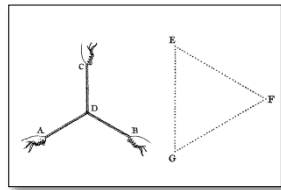
Module IV

Algebraic Graph Statics

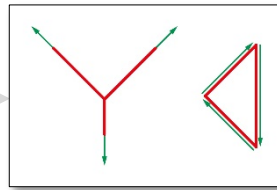


Module V

Thrust Network Analysis

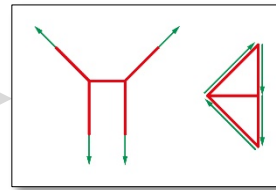


Introduction



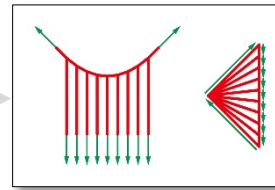
Module I

Procedural Graphic Statics I (single node)



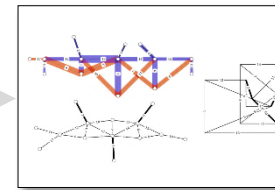
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Procedural Graphic Statics II (multi node)



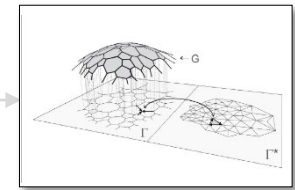
Module III

Programming Graphic Statics



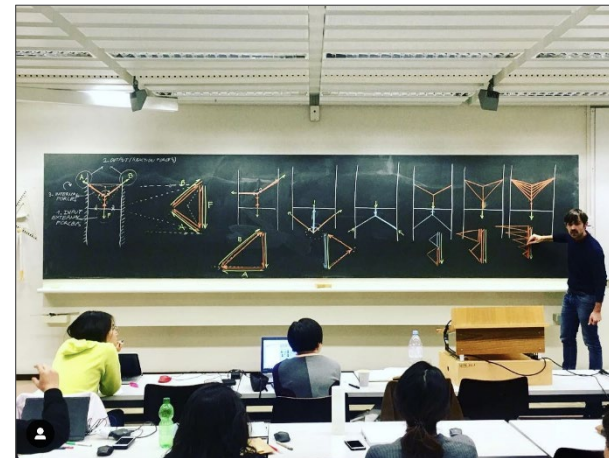
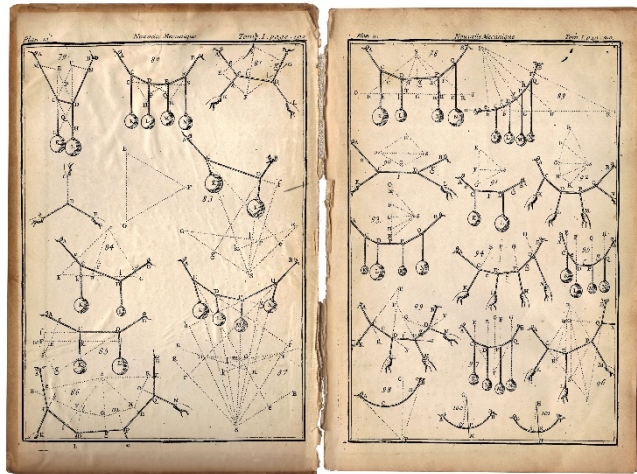
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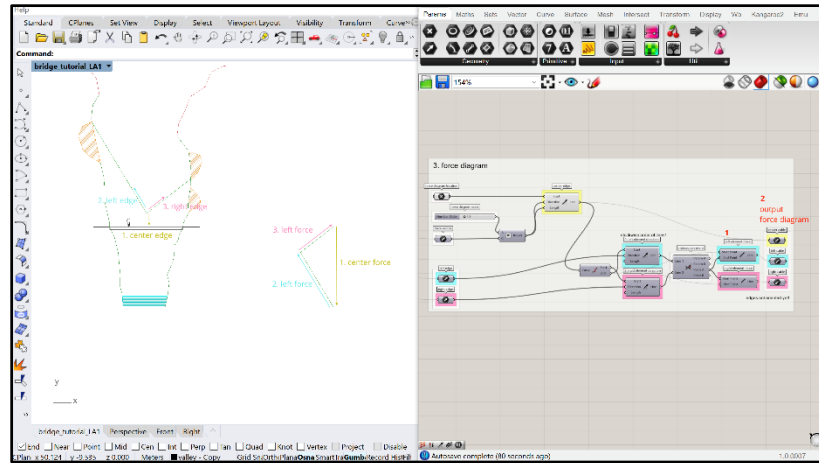
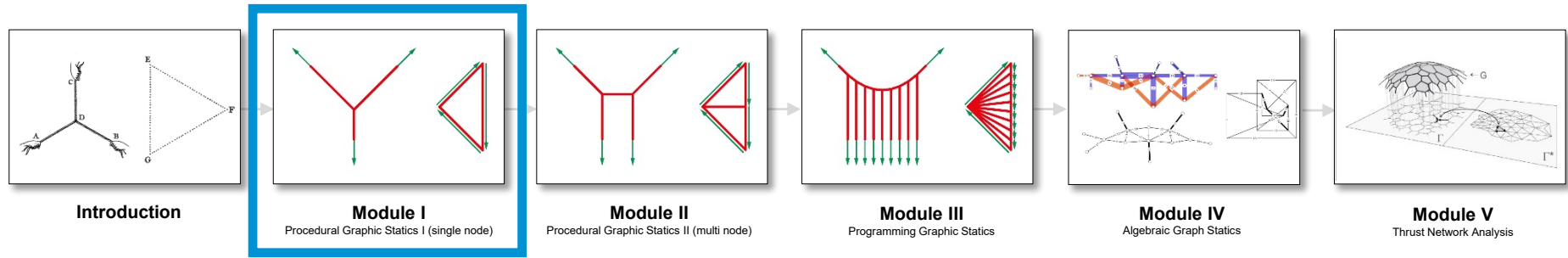
Algebraic Graph Statics



Module V

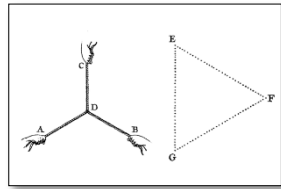
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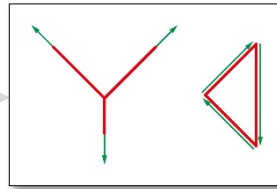


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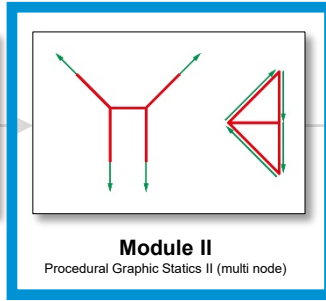




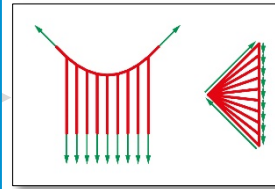
Introduction



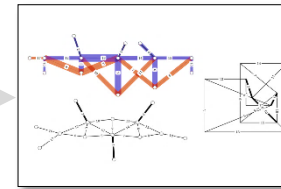
Module I
Procedural Graphic Statics I (single node)



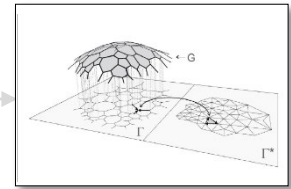
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Procedural Graphic Statics II (multi node)



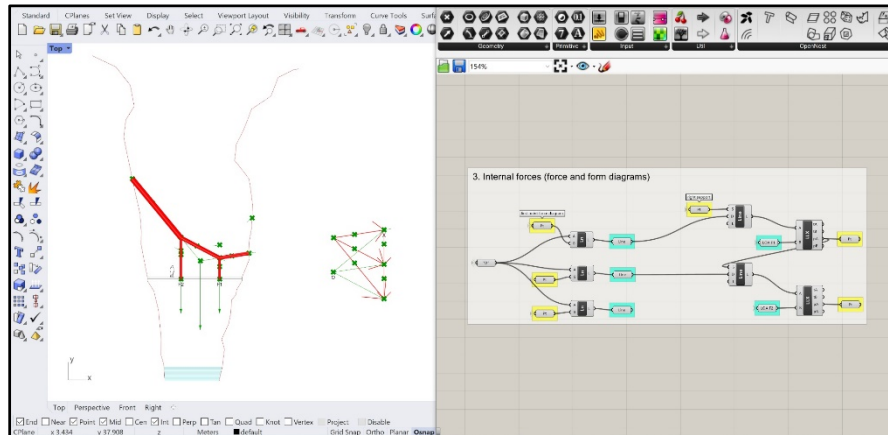
Module III
Programming Graphic Statics



Module IV
Algebraic Graph Statics

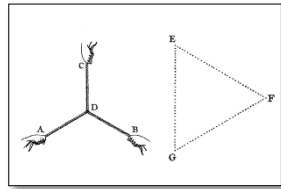


Module V
Thrust Network Analysis

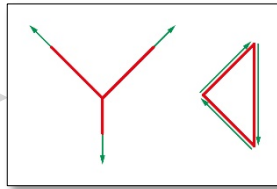


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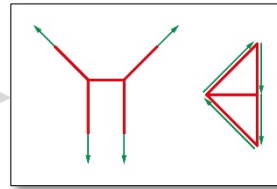


Introduction



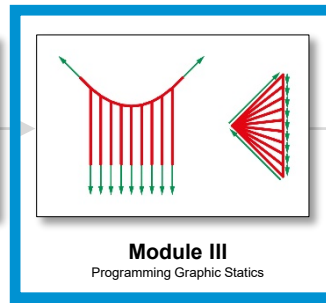
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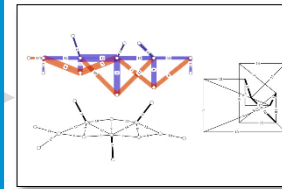
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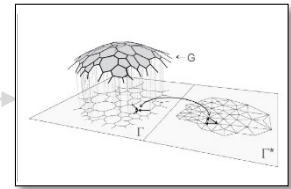
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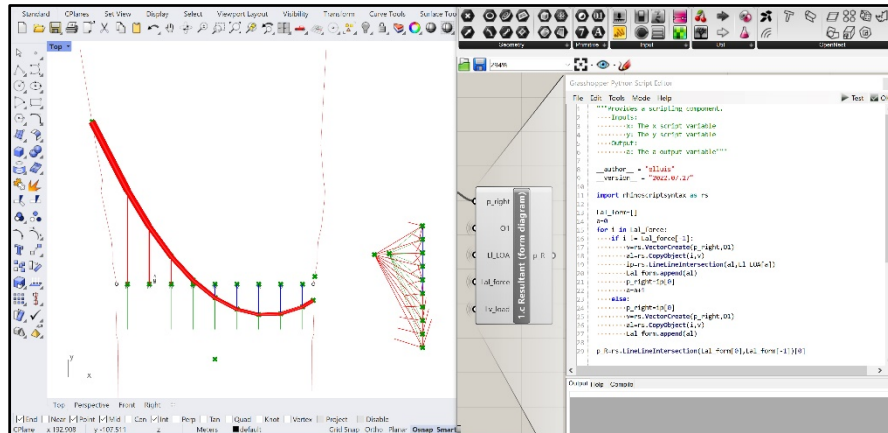
Module IV

Algebraic Graph Statics



Module V

Thrust Network Analysis

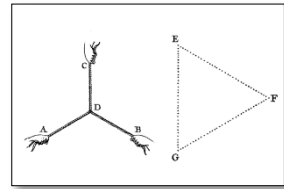


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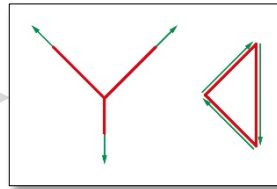


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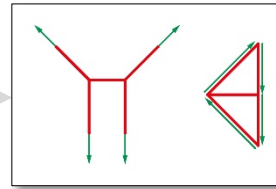




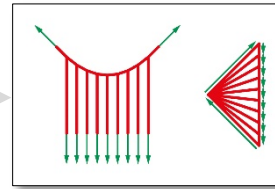
Introduction



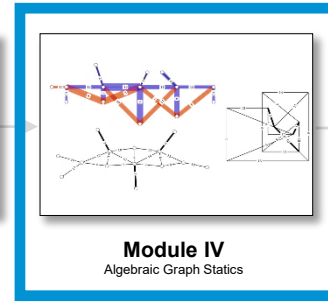
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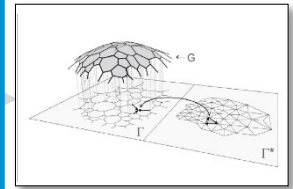
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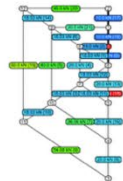
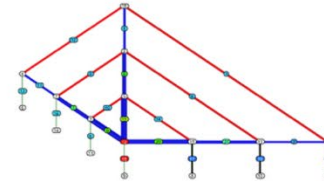
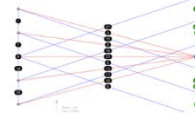
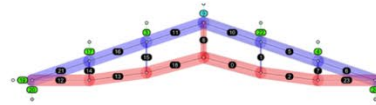
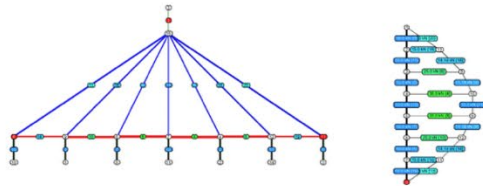
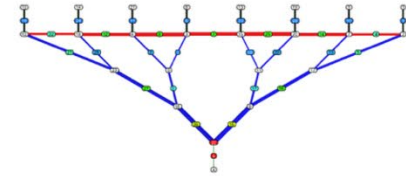
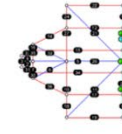
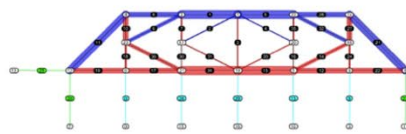
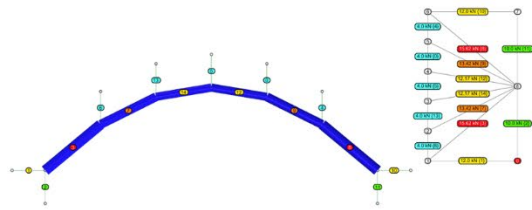
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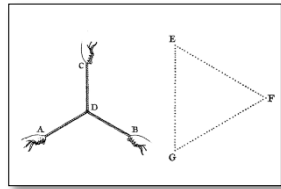


Module IV
Algebraic Graph Statics

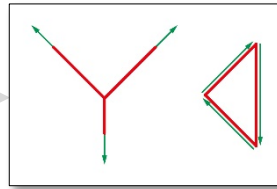


Module V
Thrust Network Analysis

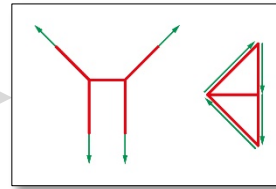




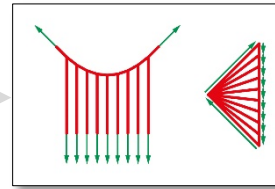
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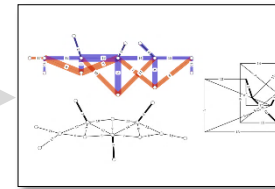
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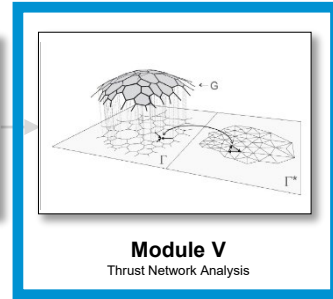
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Procedural Graphic Statics II (multi node)



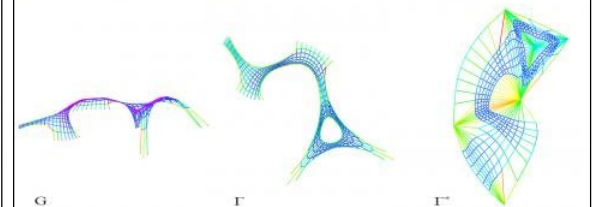
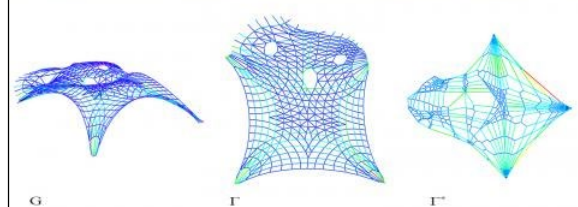
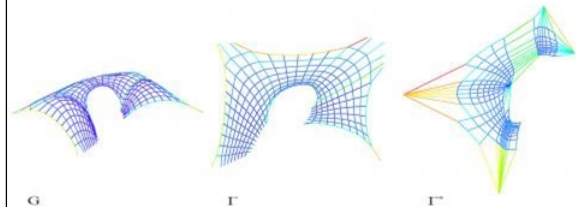
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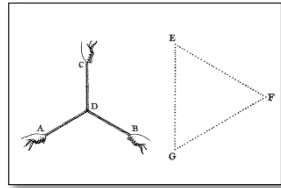


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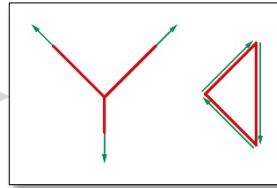


Module V
Thrust Network Analysis



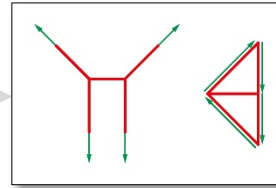


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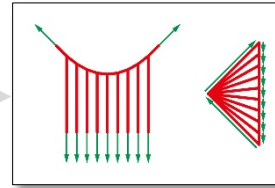
Module I

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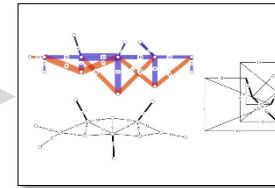
Module II

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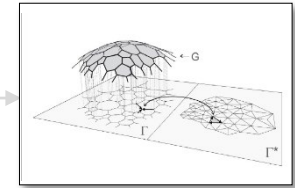
Module III

Programming Graphic Statics



Module IV

Algebraic Graph Statics



Module V

Thrust Network Analysis



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At the conclusion of this course, you will have:

- Fundamental knowledge of graphic statics.
- Basic parametric thinking and modelling with Rhinoceros and Grasshopper.
- Elementary skills in Python scripting.
- Experience using computational tools based on graphic statics for the design and analysis of structures.
- Understanding of the potential of combining graphic statics and computational tools.

Typology	Topic	Module	Week		Topic	Lead
cables & arches	Introduction		Week 1 (23/9)	Lecture	Introduction to graphic statics	Dr. Lluís Enrique
				Tutorial	Introduction to Rhino and Grasshopper	
				Work session	graphic statics + Grasshopper	
	Procedural GS (Grasshopper)	I	Week 2 (30/9)	Lecture	Algorithmic design & thinking	Dr. Lluís Enrique
				Tutorial	Single node bridge in Grasshopper	
			Week 3 (7/10)	Work session	Single node bridge in Grasshopper	
		II	Week 4 (14/10)		Quick recap of last week's exercise	Dr. Lluís Enrique
				Lecture	Computational graphic statics	
			Week 5 (21/10)	Work session	Multi-node bridge in Grasshopper	
Week 6: Seminar week						
cables & arches	Procedural GS (Python)	III	Week 7 (4/11)	Lecture	Programming	Dr. Lluís Enrique
				Tutorial	Multi-node bridge in Python	
			Week 8 (11/11)	Work session	Multi-node bridge in Python	
trusses	AGS	IV	Week 9 (18/11)	Lecture	Algebraic Graph Statics (AGS)	Dr. Lluís Enrique
				Tutorial	Interactive Graphic Statics (IGS)	Chaoyu Du
			Week 10 (25/11)	Work session	Truss analysis using IGS	
shells	TNA	V	Week 11 (02/12)	Lecture	Thrust Network Analysis (TNA)	Dr. Lluís Enrique
				Tutorial	RhinoVault 2 (rv2)	Selina Bitting
			Week 12 (09/12)	Work session	Shell design using rv2	

063-0605-22L (CSDI) - 2022
Computational Graphic Statics

Block Research Group eEQUILIBRIUM COMPAS

Q Search...

About

Syllabus

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COURSE MATERIAL

Introduction

Lecture

Tutorial

Exercise

I. Procedural GS I

II. Procedural GS II

III. Programming GS

IV. Algebraic GS

V. RhinoVAULT 2

APPENDIX

References

Slack

About



Welcome to the course Computational Structural Design I: Computational Graphic Statics!

This course presents new structural design opportunities that emerge when graphic statics, an intuitive equilibrium-based method for the form-finding, analysis and design of structures, is combined with computational tools. After a review of graphic statics fundamentals and an introduction to basic parametric tools, the course focuses on studying different computational approaches to program graphic statics algorithms for the form-finding and analysis of structures, from linear (procedural) algorithms to programs with more complex data structures. More specifically, the students will build linear parametric graphic statics models using Grasshopper, read and code basic graphic statics algorithms using Python, and explore the design space offered by tools developed by the Block Research Group such as Interactive Graphic Statics (IGS) and RhinoVault2 (rV2). The practical potential and relevance of these methods will be demonstrated through various design-oriented tutorials and exercises.

General information

- **Name:** 063-0605-22L : Computational Structural Design I (ETH course catalogue)
- **Semester:** Autumn Semester 2022
- **Lecturer:** Dr. Lluis Enríque
- **Date/time:** Fridays, 09:45 - 12:00
- **Location:** HPT C 103

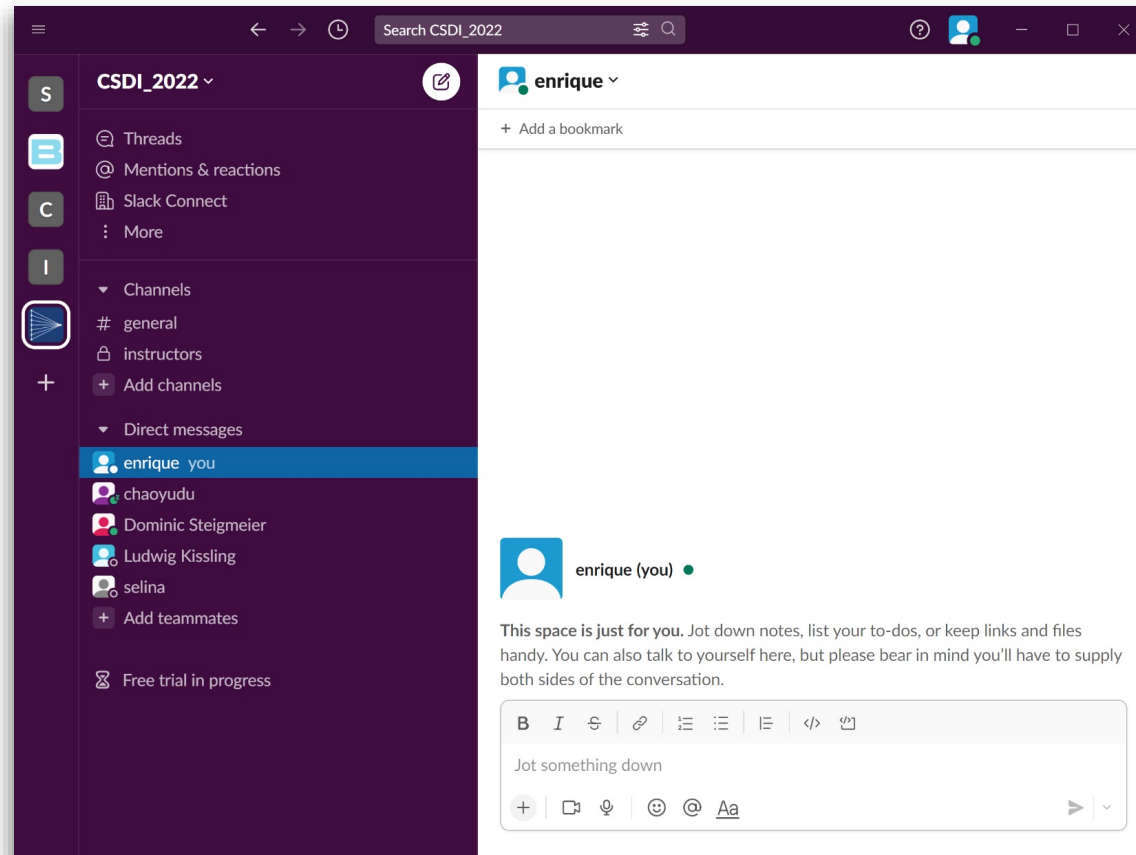
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OUTLINE

Welcome to the course Computational...

General information

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Week 1 Friday, September 23rd

9:45 – 10:30 Welcome & introduction
Lecture: Introduction to graphic statics

10:30 – 10:45 Break

10:45 – 11:30 **Tutorial: Introduction to Rhino & Grasshopper**

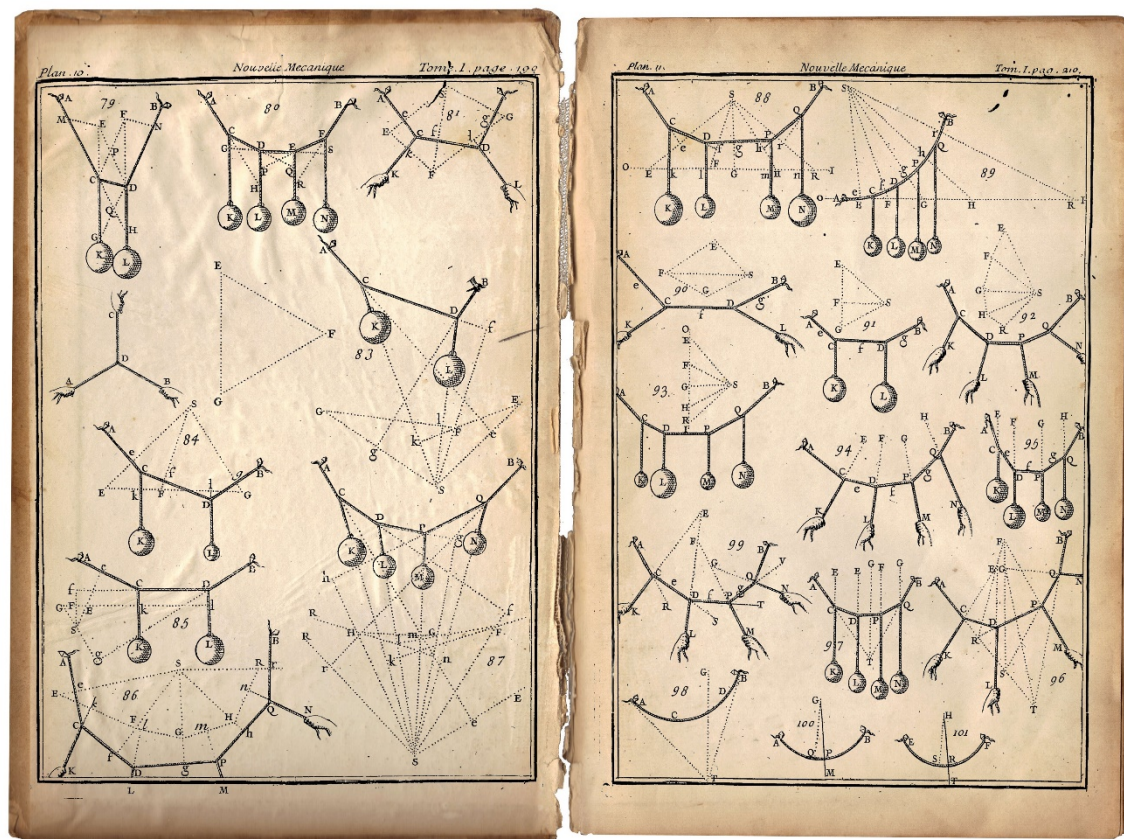
11:30 – 11:45 Break

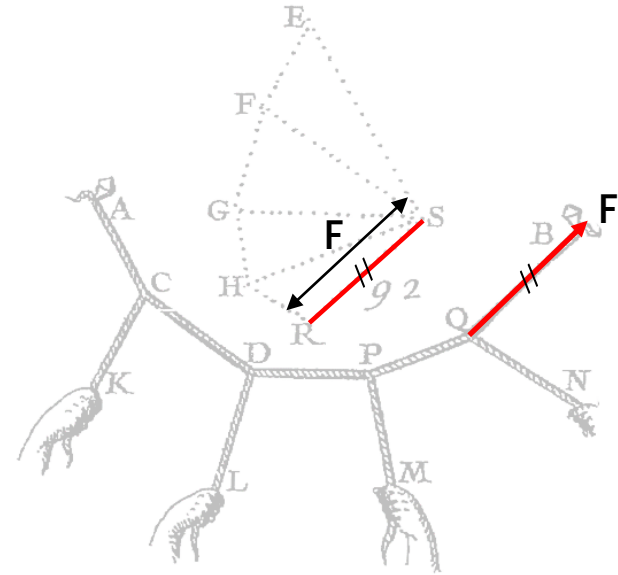
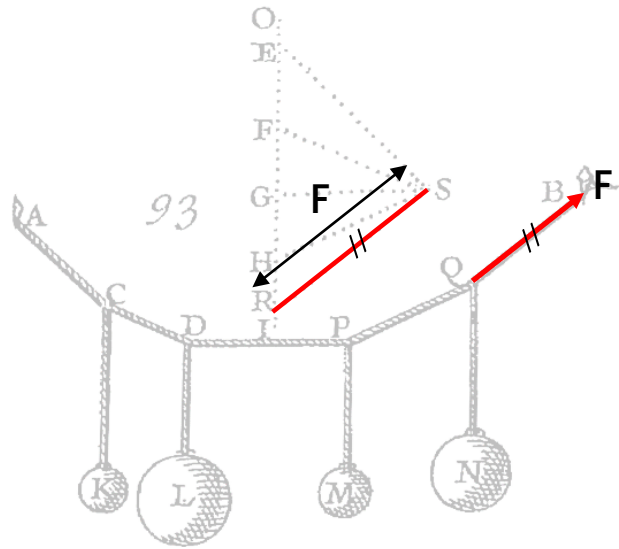
11:45 – 12:30 Exercise introduction
Work session

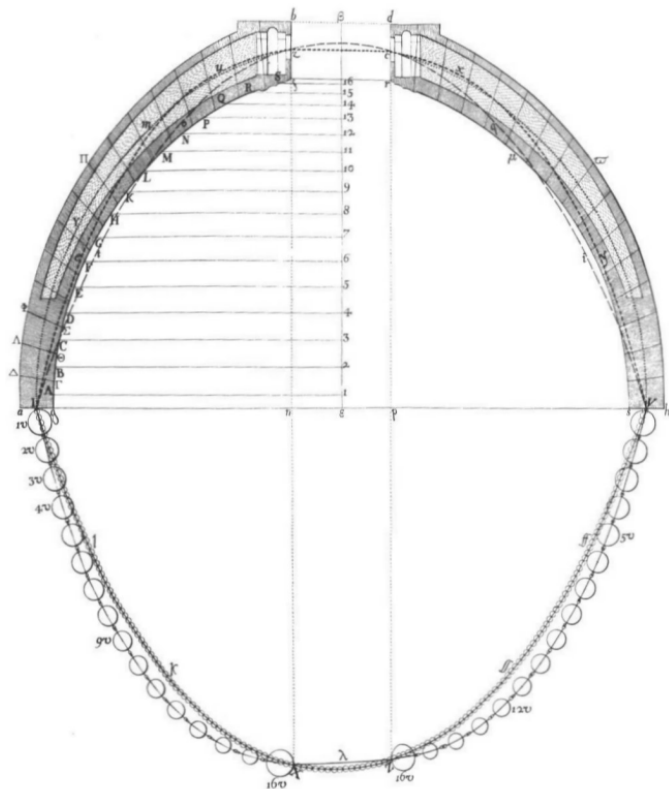
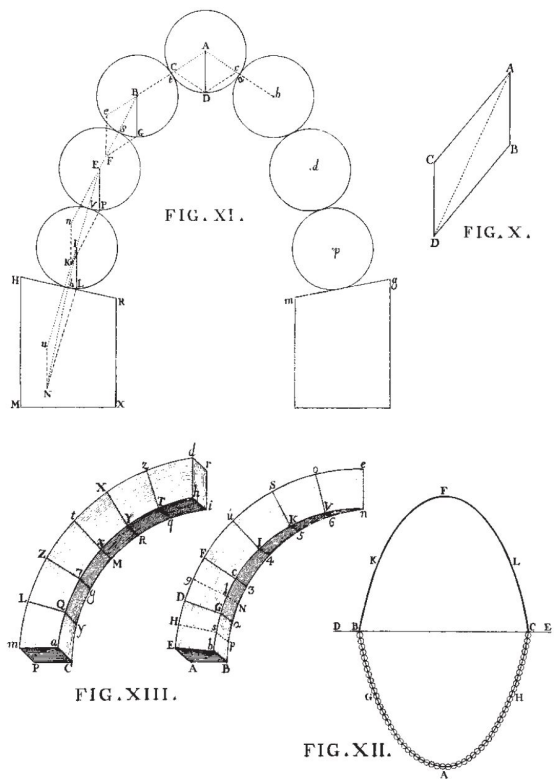
Introduction to Graphic Statics

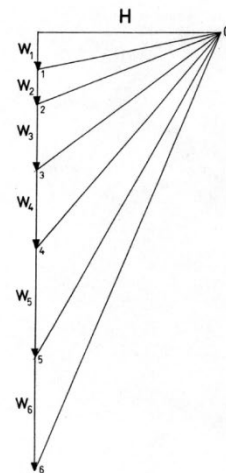
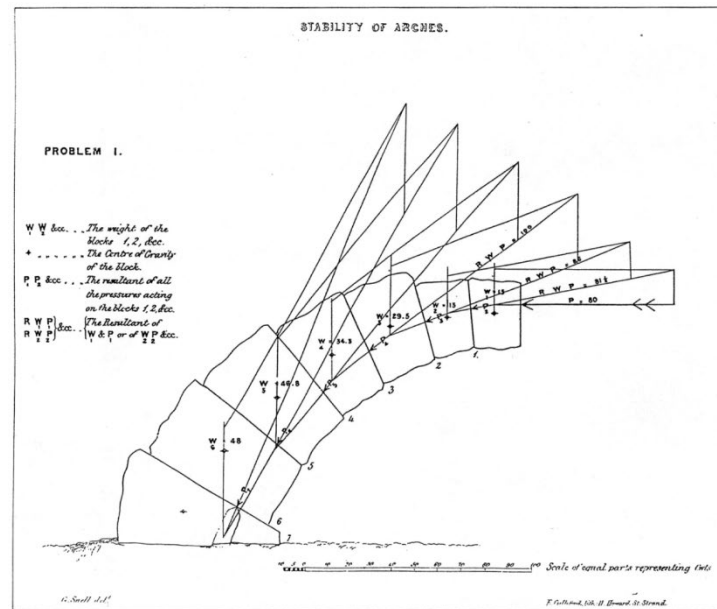
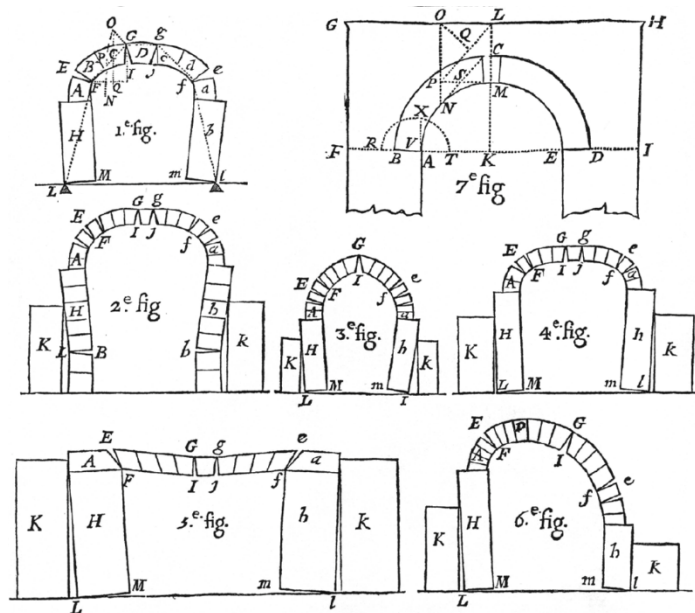
Friday, September 23rd, 2022

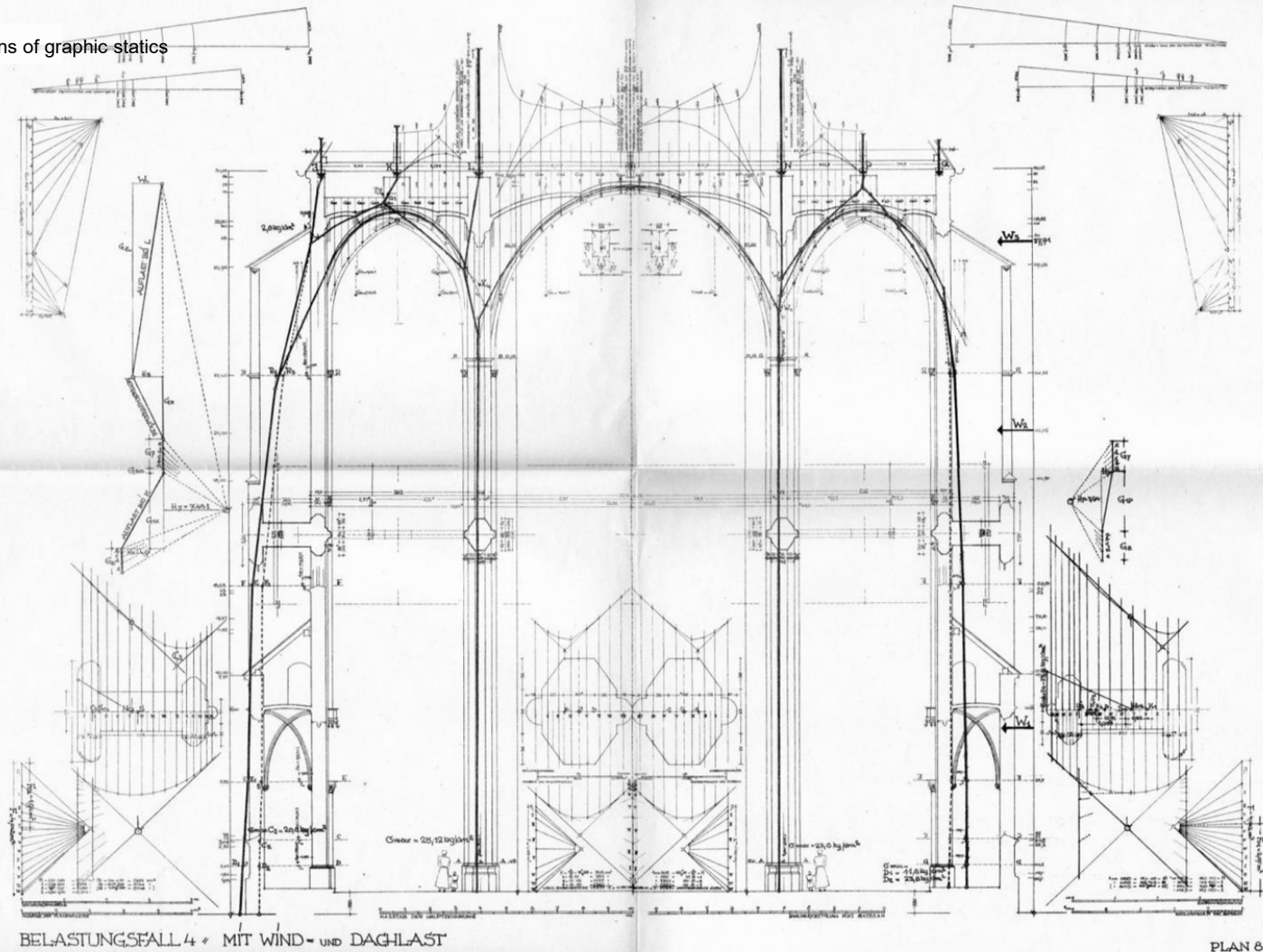
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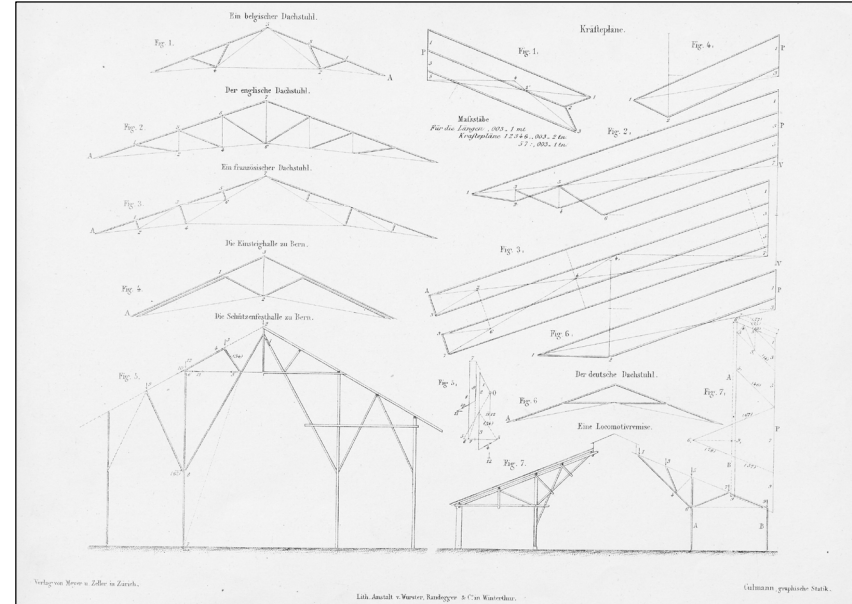
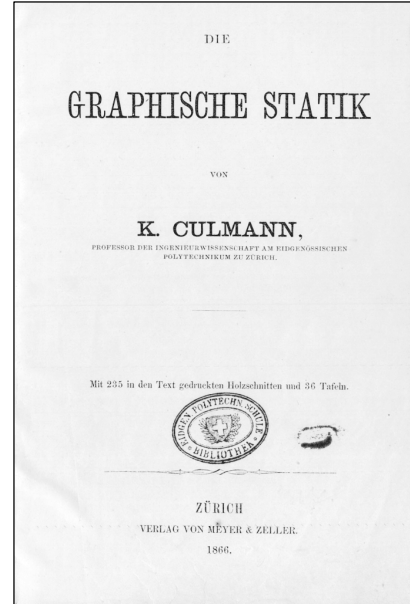












156 EXAMPLES OF FRAME- AND STRESS- DIAGRAMS. [48-

the upper extremity of the segment 1 meets the upper extremity of the segment 2; and the straight line 18 through the point (16, 1), which is both the lower extremity of the segment 16 and that of the segment 1.



Fig. 8a.

Pass on to the joint (2, 17, 19, 30). Draw 19 through the point (17, 18), and 20 through the point (2, 3), the lower end of 2 and upper extremity of 3; and we obtain the polygon 2, 17, 19, 20, which is a rectangle.

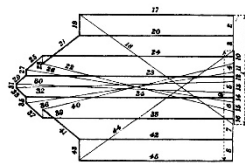


Fig. 8b.

Construct the polygon corresponding to the joint (16, 18, 19, 21, 22). For this purpose draw 21 through the point (19, 20), and 22 through the point (16, 18); we thus obtain a crossed pentagon. Continue to deal in the same manner with each of the points of application of the forces 3, 15, 4, 14, 13, 6, 12, 6, 11, 7, 10, 9, taken in succession.

Since the diagram *a*, which represents the skeleton of the structure and all the external forces, has for its axis of symmetry the vertical which passes through the centre of the figure, the diagram *b* has for its axis of symmetry the median horizontal line. For example, the triangle 9, 45, 44 is sym-

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metrical to the triangle 1, 17, 18; the rectangle 8, 45, 43, 42 to the rectangle 2, 17, 19, 20; and so on.

All the upper bars are in compression, and all the lower ones are in tension.

The diagonals and contra-diagonals are all in compression; finally two of the verticals 23, 39 are in tension, and all the rest in compression.

44. Figure 9a* represents one half of a locomotive shed. The external forces are the weights 1, 2, 3, 4, 5 applied at

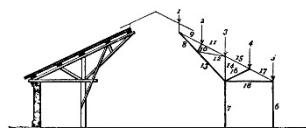


Fig. 9a.

the upper joints of the frame, and the reactions 6 and 7 of the wall and column. Again, all the external forces are parallel, and consequently the polygon of forces reduces, in diagram *b*, to one straight line. The force 8 (taken in the opposite sense to that in which it really acts) is equal to a certain part of the weight 5; by adding the difference to the other weights we get the magnitude of the force 7.

In the diagram *b* the direction of the lines 8 and 13 coincide; the first is a part of the second. Here then we have for the polygon corresponding to the joint (8, 10, 12, 13) one of those degenerate forms about which we spoke in Art. 33; the polygon is in fact a quadrilateral 8, 10, 12, 13, having three of its vertices (13, 9), (9, 10), (12, 13) in one straight line.

The polygon 5, 17, 18, 6, corresponding to the point where

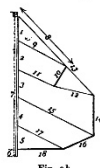


Fig. 9b.

* This example is taken from PLATE of the *Atlas of Graphical Statics* of CREMONA, 1st edition. As previously stated, the two diagrams are not rigorously reciprocal.

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the roof is supported by the wall, presents an analogous degenerate form, since the vertices (6, 5), (upper point of the segment (6), (5, 17), and (18, 6) all lie in the same straight line.

The lower bars 8, 13, 18 are in compression, as well as the diagonals 10, 14, 16, the column 7 and the wall 6; while the upper pieces 9, 11, 15, 17 and the diagonal 12 are in tension.

45. Diagram *a* of Fig. 10 represents a truss at the upper joints of which are applied the oblique forces 1, 2, ..., 7, which

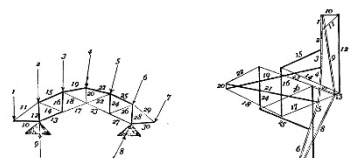


Fig. 10a.

may be considered as the resultants of the dead-loads and wind pressure; the forces 8, 9 represent the reactions at the supports.

The polygon of external forces is drawn in diagram *b* with double lines.

We construct successively the triangle 1, 10, 11, the quadrilateral 9, 10, 12, 13, the pentagon 2, 11, 12, 14, 15, the quadrilateral 13, 14, 16, 17, the crossed pentagon 3, 15, 16, 18, 19; the crossed quadrilateral 4, 19, 20, 21, the pentagon 17, 18, 20, 22, 23, and so on.

The upper bars 16, 19, 21, 23 are in compression, as well as the lower bars 10, 13, 20, and the verticals 12, 16, 24, 28; whilst all the remaining bars of the structure are in tension.

46. The diagram *a* of Fig. 11 represents a suspension bridge, loaded at each of its upper joints with weights 1, 2, ..., 8, and at each of its lower joints with weights 10, 11, 21, ..., 16; the weights are kept in equilibrium by the two

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oblique reactions 9, 17 at the two extreme points of the structure*.

The polygon of external forces has its first eight sides in succession along the same vertical straight line, and its seven

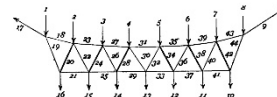


Fig. 11a.

last sides situated in another vertical straight line. The oblique sides 9 and 17 intersect, so that the polygon is a crossed one. We construct successively the polygons 1, 17, 19, 18; 16, 19, 20, 21; 2, 18, 20, 22, 23; 15, 21, 22, 24, 25; 3, 23, 24, 26, 27; and so on; most of which are crossed.

Diagram *b* shows that the upper bars are all in tension, and that the tension decreases from the ends towards the

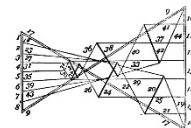


Fig. 11b.

middle of the structure; the bars of the lower boom are also all in tension, but in them the tension decreases from the middle towards the ends.

The extreme diagonals and contra-diagonals are in tension; in the portion situated to the left of the axis of symmetry, the diagonals or braces are alternately in tension and compression; similarly they are on the right but in the reverse order. Considering separately the ties and struts, we see that the internal

* This example is analogous to one of those studied by Maxwell in his memoir of 1870.

PLATE VII.

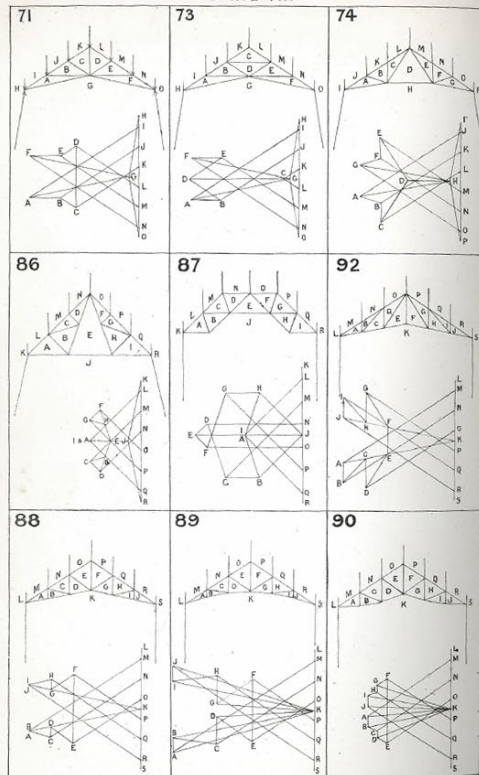
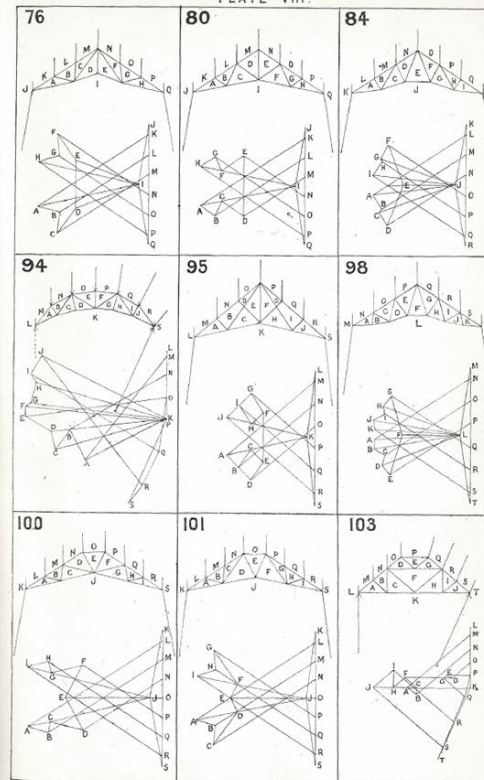
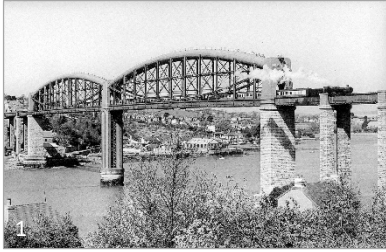


PLATE VIII.





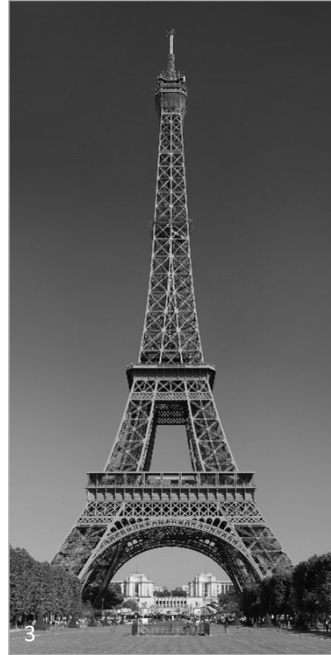
Royal Albert Bridge (1859)

Isambard Kingdom Brunel



Garabit Viaduct (1884)

Maurice Koechlin



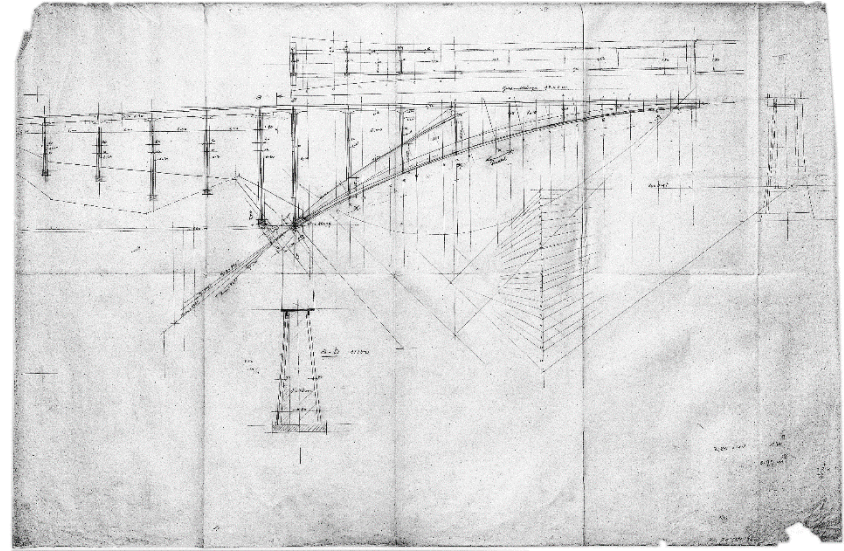
Eiffel Tower (1887)

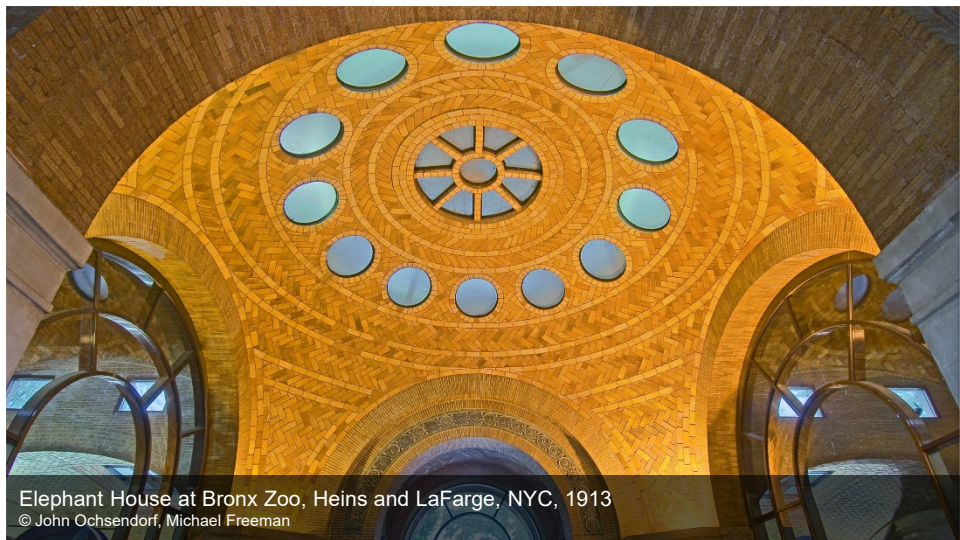
Maurice Koechlin, Émile Nouguier

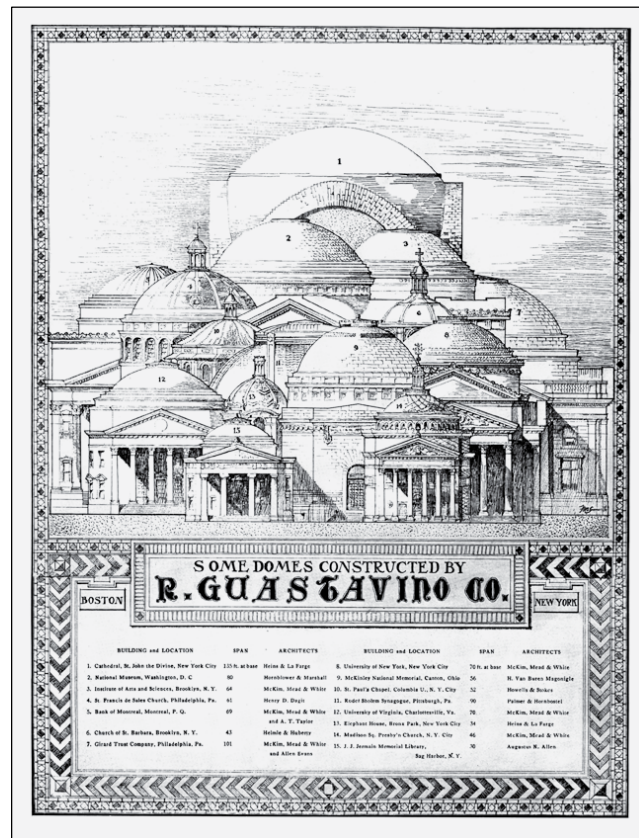
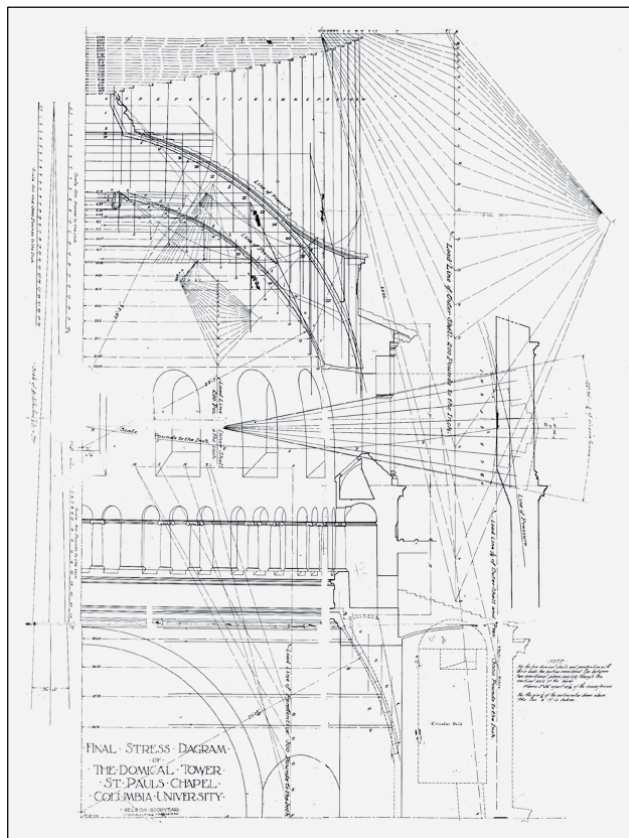


Forth Bridge (1890)

Sir John Fowler, Sir Benjamin Baker









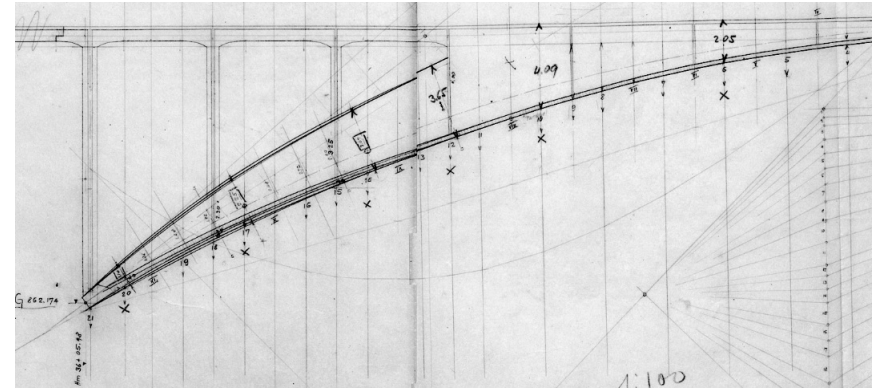
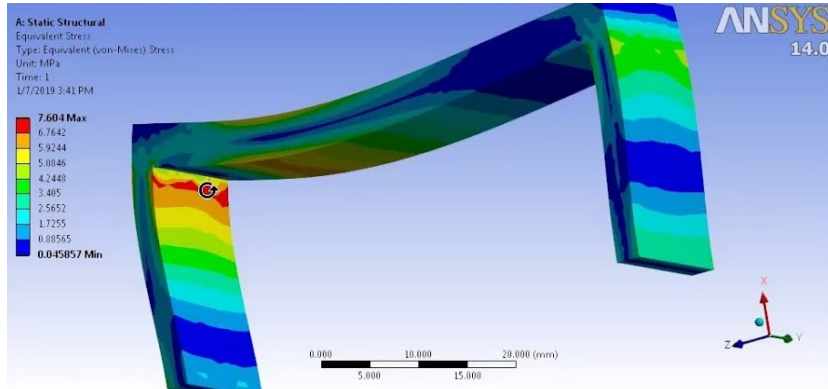
TOTAL LOAD ON
ONE HANGER

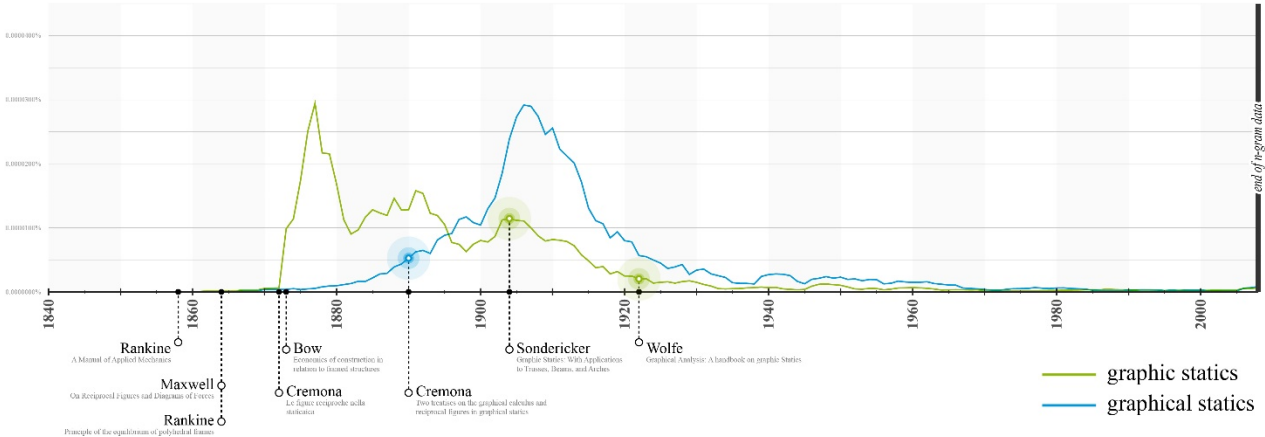
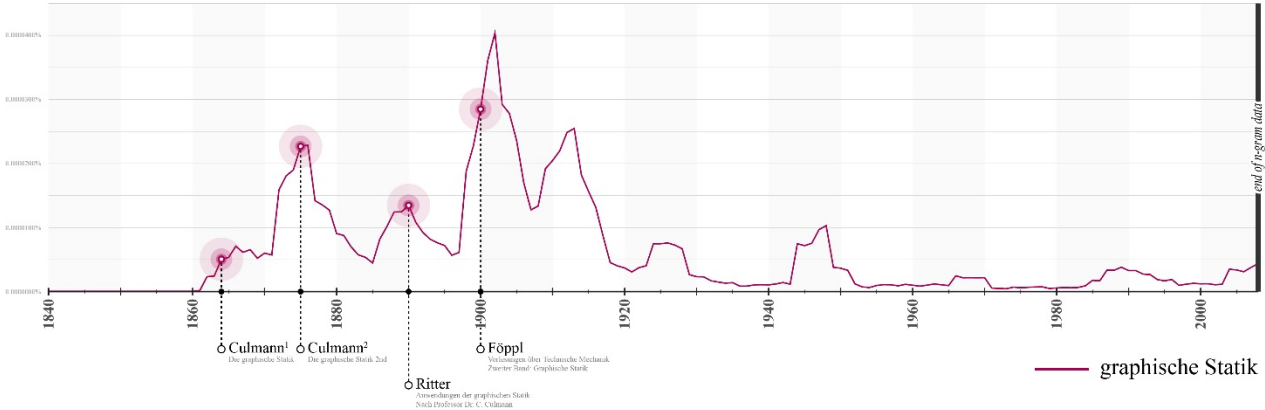
LOAD = 60 psf

$P = 60 \text{ lb} (400 \text{ ft})$

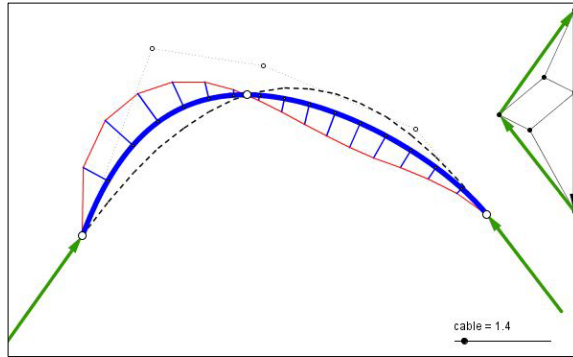
$P = 24,000 \text{ lb}$
 $= 24 \text{ K}$





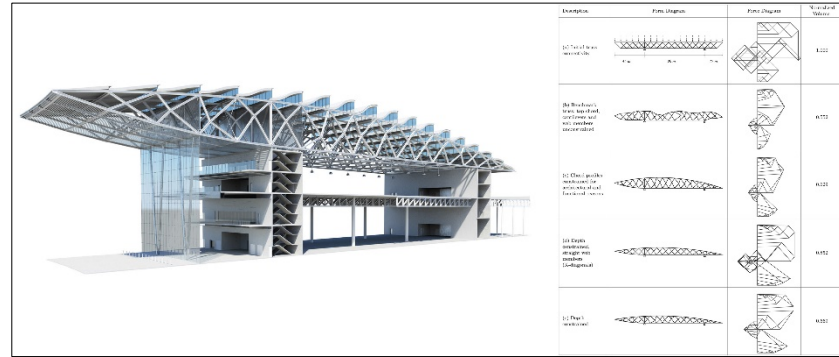






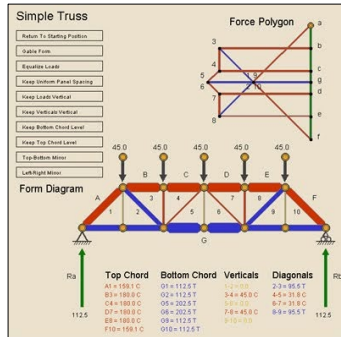
eEQUILIBRIUM

Van Mele et al. (2009-)



Structural Optimization using Graphic Statics

Beghini, et al. & SOM (2013)



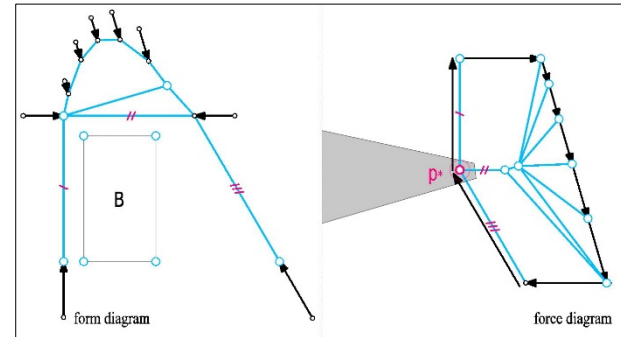
Active Statics

Greenwood & Allen (2003)



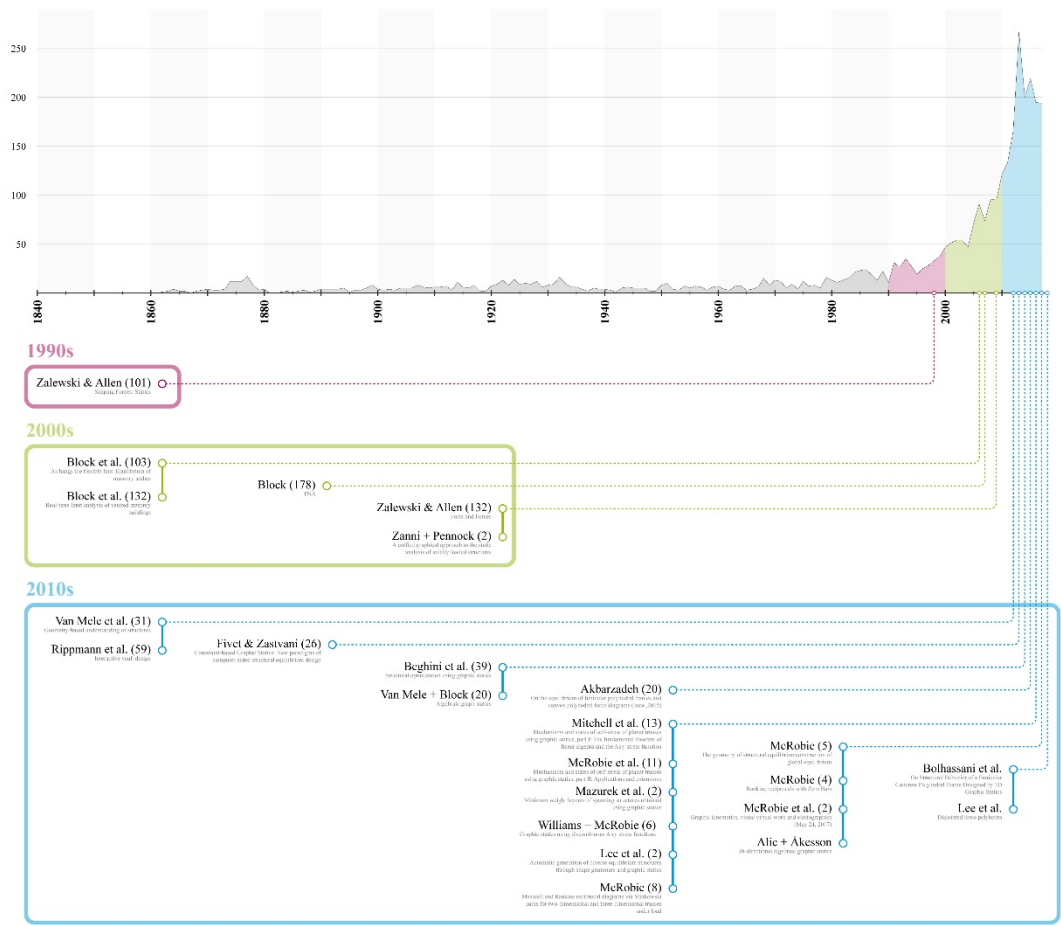
Form and Forces

Allen & Zalewski (2009)

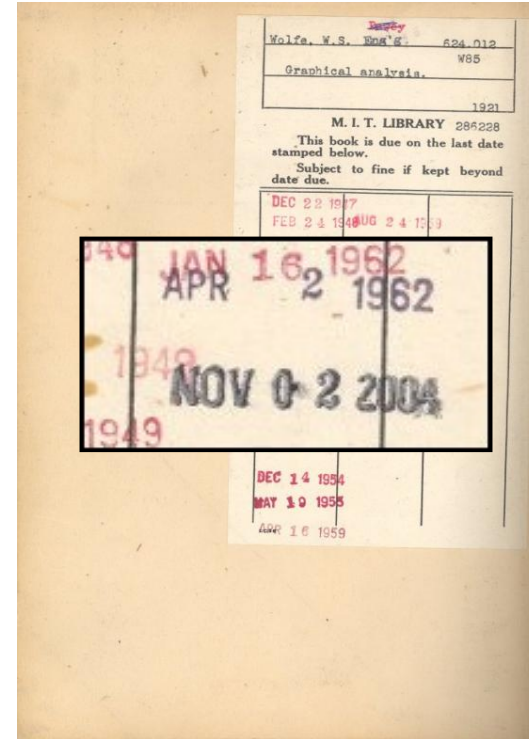
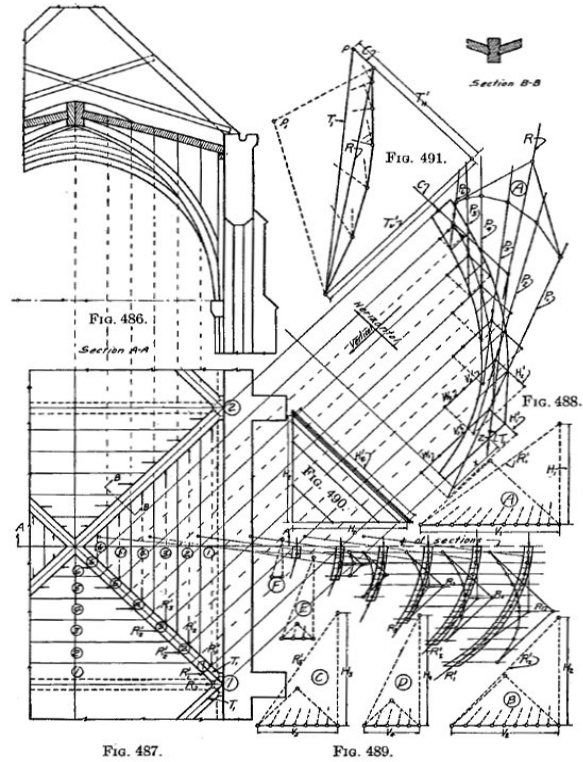


Constraint-based Graphic Statics

Fivet & Zastavni (2013)



Timeline of graphic statics publications (early 2018) | from PhD dissertation by Lee (2018)









PROFESSOR DR
CARL CULMANN
VON 1855 BIS 1881
VORSTAND DER
INGENIEURSCHULE
DES EIDGENÖSSISCHEN
POLYTECHNIKUMS



Karl Culmann

- Founder of graphic statics: “Die graphische Statik” (1864-1866)
- First Professor of Civil Engineering at the ETH Zurich (1855-1883)
- Director of the Polytechnic (1872-1875)



Karl Wilhelm Ritter (ETHZ diploma, 1868)

- Professor of graphic statics and bridge construction (appointed in 1882)
- Director of the Polytechnic (1887-1891)
- Extended applications of Culmann’s graphic statics:
“Anwendungen der graphischen Statik nach Professor Dr. Culmann” (1888-1906)



Robert Maillart
(ETH diploma, 1894)



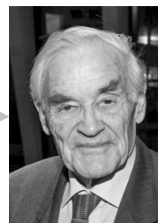
Othmar Ammann
(ETH diploma, 1902)



Pierre Lardy
(ETH diploma, 1928)



Heinz Isler
(ETH diploma, 1950)



Christian Menn
(ETH diploma, 1950)



Joseph Schwartz
(ETH professor, 2008-)



Philippe Block
(ETH professor, 2009-)

